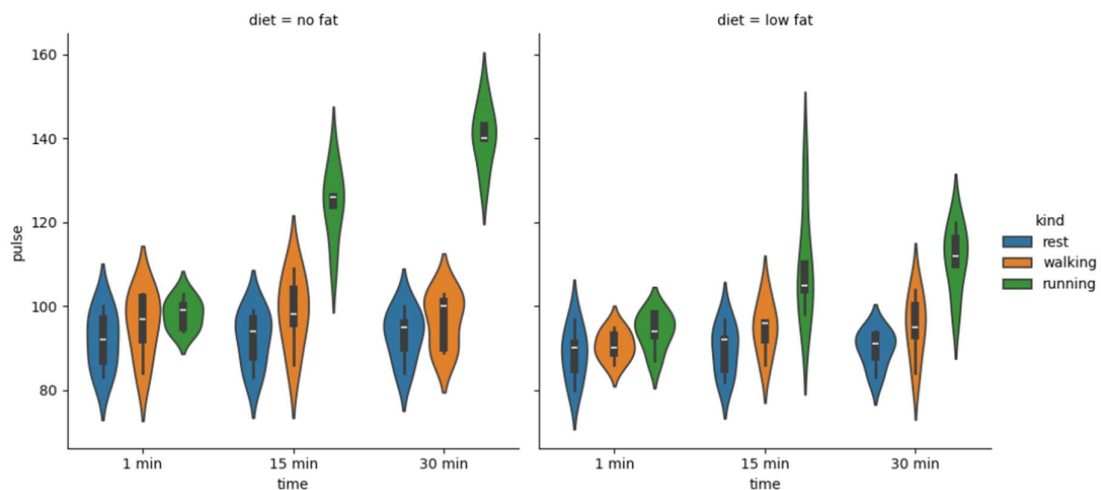


Factor Plot

A **factor plot** (now officially known as `catplot` in the seaborn library) is a data visualization tool used to show the relationship between a numerical variable and one or more categorical variables. It allows you to display multiple types of plots—such as bar, box, violin, or point plots—across different subsets of data

Depending on your data, a factor plot can be configured using the `kind` parameter

```
#catplot
import pandas as pd
import seaborn as sns
from matplotlib import pyplot as plt
df = sns.load_dataset('exercise')
sns.catplot(x = "time", y = "pulse", hue = "kind", kind = 'violin', col = "diet", data = df);
plt.show()
```



The outer shape shows the probability density of the data. Wide sections represent a higher concentration of data points, while narrow sections show lower density. The inside of the violin usually contains a miniature box plot. The white dot indicates the median, the thick central bar represents the interquartile range, and the thin line shows the rest of the distribution.